

The pooling factor predicts cross-validation reliability: a closed-form triage and Rao–Blackwellised cure for hierarchical LOO

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Abstract

For hierarchical models, Pareto-smoothed importance-sampling LOO (PSIS-LOO) fails on exactly the folds where a random-effect coordinate is data-driven and its group is small. We show that the Gelman–Pardoe pooling factor and the associated structural leverage predict these folds from the model structure and group sizes, without forming importance weights. In Gaussian linear mixed models the leverage reduces to group size, giving a design-time map that separates the $\hat{k}_i > 0.7$ folds with AUC 0.96; across replicated logistic GLMMs the post-fit, weight-free predictor reaches AUC 0.81. The cure for a flagged fold is integrated importance sampling (marginalise the random-effect block, importance-sample only the base parameters), the standing recommendation of the `loo` documentation and available in closed or quadrature form in the latent Gaussian literature [Vehtari et al., 2016, Merkle et al., 2019, Bürkner et al., 2021]. We contribute its observation-level specialisation for random-intercept GLMMs, an analytic Gaussian downdate and a 1-D quadrature for Bernoulli, binomial and Poisson responses, packaged as a drop-in `rb_loo(fit)` for the Stan/`brms`/`loo` ecosystem. Against exact refits, this marginalised estimator (RB-LOO) is $3\times$ more accurate than moment matching on singleton-heavy logistic GLMMs. On an overdispersed count dataset with 97 failing folds, moment matching leaves 37 uncorrected and is no more accurate than raw PSIS-LOO, while RB-LOO reproduces the 82-minute exact refit (elpd RMSE 0.04) at no cost. The error changes decisions: comparing this model against a negative-binomial alternative, PSIS-LOO reports decisive evidence ($z = 4.9$) and `relloo` at the recommended threshold reports significant evidence ($z = 3.4$) for the more complex model, where an exact fold-wise analysis, reproduced by RB-LOO, finds the two indistinguishable ($z = 1.0$). A base–fiber decomposition explains the split: a Schur complement separates case-deletion influence into a vertical (pooling) term, which governs where PSIS-LOO fails, and a horizontal (variance-component) term, which governs where RB-LOO’s own residual importance sampling is strained. The resulting two-level triage (pooling factor, then residual base- $\hat{k}$) recovers the exact answer while refitting only the few folds that need it. The marginalisation is not new; the contributions are the a-priori pooling-factor triage, the packaged closed forms, the head-to-head showing marginalisation dominates the importance-sampling repairs, and the geometric account.

Keywords: cross-validation; leave-one-out; PSIS; hierarchical models; partial pooling; Rao–Blackwellisation; importance sampling; pooling factor.

1 Introduction

Leave-one-out cross-validation (LOO-CV) is a standard tool for Bayesian model assessment, and Pareto-smoothed importance sampling [PSIS-LOO; Vehtari et al., 2017, 2024] has made it cheap enough for routine use: a single model fit yields an estimate of the expected log predictive density (elpd) for every held-out observation, together with a per-fold diagnostic $\hat{k}_i$ that flags when the importance-sampling approximation is unreliable. The diagnostic is a safety valve: when $\hat{k}_i > 0.7$, the recommendation is to refit the model with observation i removed [Vehtari et al., 2017].

For hierarchical models this safety valve fires often, and in a structured way. When an observation belongs to a small group (a singleton, in the limit), deleting it removes most of the information about that group’s random effect, so the leave-one-out posterior of that coordinate moves far from the full-data posterior, the importance weights acquire a heavy tail, and $\hat{k}_i$ crosses the threshold. The recommended cure, an exact refit, is what PSIS-LOO was meant to avoid; the cheaper alternative, moment matching [Paananen et al., 2021], still requires a per-fold optimisation.

This paper makes three points, validated on replicated simulations and real data and packaged as a drop-in tool.

C1 (predict). The pooling factor of Gelman and Pardoe [2006], the fraction of a group’s posterior precision contributed by the prior, and the associated per-observation structural leverage predict which folds PSIS-LOO fails on, without forming importance weights. In a Gaussian linear mixed model the per-observation Fisher information is constant, so the leverage reduces to group size and the predictor is a design-time map, available before the data are seen; it separates the $\hat{k}_i > 0.7$ folds with AUC 0.96. For GLMMs the predictor uses the fitted variance component and Fisher weights, so it requires one model fit, but still no importance weights and no case deletion, and reaches AUC 0.81 on replicated logistic models with heterogeneous group sizes. We are not aware of prior work mapping the pooling factor to PSIS-LOO reliability.

C2 (cure), and its provenance. The cure for a flagged fold is to integrate out the random-effect block and importance-sample only the base parameters. The idea is not ours: it is integrated importance-sampling LOO, the marginal-likelihood LOO of Merkle et al. [2019], the latent-marginalised LOO with quadrature of Vehtari et al. [2016], the closed forms of Bürkner et al. [2021], and the standing recommendation of the `loo` documentation for hierarchical PSIS failures. Marginalising the deleted coordinate, rather than reweighting it, removes the heavy-tailed weight. We contribute the closed-form observation-level specialisation for random-intercept GLMMs (an analytic Gaussian rank-1 downdate; a 1-D quadrature for Bernoulli, binomial and Poisson), packaged as a drop-in `rb_loo(fit)`, and the head-to-head that the C1 flag makes possible: on the folds that matter the marginalised estimator matches exact refits at zero optimisation cost, is $3\times$ more accurate than moment matching [Paananen et al., 2021] on singleton-heavy logistic GLMMs, and corrects failures that moment matching leaves in place on a real count dataset. We call the estimator RB-LOO (Rao–Blackwellised LOO): integrated IS is Rao–Blackwellisation of the LOO estimator (Section 3), so the name describes a standard operation.

C3 (triage and explain). A base–fiber bundle structure [Bindoff, 2026] organises the two failure modes. A Schur complement (Section 3) splits each observation’s case-deletion influence into a vertical (pooling) term, which governs where PSIS-LOO fails and which C1 exploits, and a horizontal (variance-component) term, which governs where RB-LOO’s residual base importance sampling is strained. This gives a two-level triage (pooling factor, then residual base- $\hat{k}$) that recovers the exact LOO answer while refitting only the few percent of folds where RB-LOO is itself unreliable.

The random-effect marginalisation (C2) is prior art. The structural leverage is classical case-influence machinery [Wei et al., 1998] in its modern hierarchical form [Lovison et al., 2026]; that influential observations produce high $\hat{k}$ is known [Peruggia, 1997, Vehtari et al., 2017]; and INLA’s cross-validatory CPO computes the same marginal LOO predictive deterministically [Rue et al., 2009, Held et al., 2010]. Our contribution is the composition: the a-priori pooling-factor-to-PSIS-reliability map, the packaged closed forms, the two-level triage, the head-to-head the flag enables, and the geometric account. Section 2 draws the boundaries.

2 Background and related work

PSIS-LOO and the $\hat{k}$ diagnostic. Write the full-data posterior as $p(\theta | y)$ and the pointwise likelihood as $p(y_i | \theta)$. PSIS-LOO estimates $\text{elpd}_i = \log p(y_i | y_{-i})$ by importance-reweighting the full-data posterior draws with weights $w_i^{(s)} \propto 1/p(y_i | \theta^{(s)})$, Pareto-smoothing the largest weights, and reporting the fitted Pareto shape $\hat{k}_i$ as a reliability diagnostic [Vehtari et al., 2017, 2024]. The importance-sampling variance is infinite when the leave-one-out posterior has heavier tails than the proposal in the relevant direction; $\hat{k}_i > 0.7$ signals this. Peruggia [1997] first characterised the case-deletion importance weight variance.

The pooling factor. For a group j with random effect α_j , prior precision $1/\sigma_u^2$ and data precision I_j (the Fisher information the group’s observations carry about α_j), the Gelman–Pardoe pooling factor is $\pi_j = (1/\sigma_u^2)/(I_j + 1/\sigma_u^2)$, the fraction of the posterior precision of α_j contributed by the prior [Gelman and Pardoe, 2006]. A singleton in a weakly-informative hierarchy has π_j near 1 (fully pooled, data-starved); a large group has π_j near 0. The per-observation *structural leverage* $h_i = I_i/(I_j + 1/\sigma_u^2)$, with within-group Fisher additivity $I_j = \sum_{i \in j} I_i$, satisfies $\sum_{i \in j} h_i = 1 - \pi_j$ [Bindoff, 2026], and is the shrinkage-weighted observation leverage of Lovison et al. [2026].

Conditional versus marginal LOO, and integrated importance sampling. A hierarchical model admits two LOO estimands [Merkle et al., 2019]: the *conditional* predictive $p(y_i | \theta, \alpha_{j(i)})$ integrated over the leave-one-out posterior, and the *marginal* (integrated) predictive $p(y_i | y_{-i})$ with the random effects marginalised. These coincide as estimands (integrating the conditional likelihood over the joint leave-one-out posterior, which includes α , gives the marginal predictive), but they differ sharply as importance-sampling targets: PSIS over the conditional likelihood reweights the full posterior in the α direction, which is exactly where the data-driven random effect makes the weights heavy-tailed, whereas marginalising α first removes that direction from the importance sampling. This *integrated importance sampling* is well established. Merkle et al. [2019] argue for the marginal predictive and note its better-behaved weights; Vehtari et al. [2016] compute LOO for latent Gaussian models by marginalising the latent value analytically (Gaussian likelihood) or by 1-D quadrature (logistic, Poisson), the same machinery this paper specialises to grouped GLMMs; Bürkner et al. [2021] give closed-form integrated LOO with a rank-1 downdate for non-factorised normal and Student- t models; and the `loo` documentation recommends integrating out group-specific parameters, “analytically or by quadrature,” precisely to avoid PSIS failure in hierarchical models. INLA computes the same marginal LOO predictive deterministically as its cross-validatory CPO [Rue et al., 2009, Held et al., 2010], and leave-group-out CV [Liu et al., 2022] targets a related group-level predictive. RB-LOO is a closed-form, triage-equipped instance of this integrated-IS family, operating post-hoc on existing MCMC draws within the `loo` ecosystem; throughout we compare only to refits that target the same (marginal) estimand, so the comparison is fair.

Cures for high $\hat{k}$. Given a high- $\hat{k}$ fold, the alternatives to a refit are all importance-sampling repairs that keep the conditional target and fix the proposal. Moment matching [Paananen et al., 2021] applies an affine transformation to the posterior draws for each high- $\hat{k}$ observation, chosen by a per-fold optimisation; mixture / implicitly-adaptive importance sampling [Silva and Zanella, 2024] builds a mixture proposal with guaranteed finite variance for the many-high- $\hat{k}$ regime. We take moment matching (in its robust split-transformation form, the recommended default) as the primary state-of-the-art comparator, and contrast the mechanism with mixture IS: these repair the proposal for the *conditional* weights, whereas RB-LOO removes the offending direction from the problem by marginalisation. Our empirical finding is that on singleton-driven folds the affine repair cannot represent a random effect reverting to its prior, so marginalisation dominates.

Hierarchical leverage and influence. The generalised leverage $g_i^\top G^{-1} g_i$ and its Schur decomposition are classical [Wei et al., 1998]; the hierarchical case is treated by Hodges and Sargent [2001], Cui et al. [2010] (who partition effective degrees of freedom and cite the pooling factor) and, most directly, Lovison et al. [2026], whose augmented hat matrix gives the subject-level structural leverage. These works hold the variance components fixed (Lovison) or measure complexity rather than influence (Cui–Hodges), and none addresses cross-validation reliability; Section 3 demarcates our claim accordingly.

3 Theory: the orthogonal decomposition of case-deletion influence

We work in the base–fiber coordinates of the hierarchical bundle [Bindoff, 2026]: base θ (hyperparameters: population effects, variance components, GP hypers), fiber α (group coordinates), with Fisher metric

$$G = \begin{pmatrix} G_{BB} & G_{BF} \\ G_{FB} & G_{FF} \end{pmatrix},$$

G_{FF} block-diagonal across groups. An observation i touches only its own group, so its score splits as $g_i = (g_i^B, g_i^F)$ with g_i^F supported on the $\alpha_{j(i)}$ block.

Proposition 1 (Orthogonal leverage identity, T1). *With $A = -G_{FF}^{-1}G_{FB}$, $\tilde{g}_i = g_i^B + A^\top g_i^F$ the horizontal lift of the score, and $M = G_{BB} - G_{BF}G_{FF}^{-1}G_{FB}$ the Schur complement (the marginal base precision),*

$$g_i^\top G^{-1} g_i = \underbrace{g_i^{F\top} G_{FF}^{-1} g_i^F}_{\text{vertical (pooling)}} + \underbrace{\tilde{g}_i^\top M^{-1} \tilde{g}_i}_{\text{horizontal (base leverage)}}.$$

Proof. The block inverse of G is $G^{-1} = \begin{pmatrix} M^{-1} & -M^{-1}G_{BF}G_{FF}^{-1} \\ -G_{FF}^{-1}G_{FB}M^{-1} & G_{FF}^{-1} + G_{FF}^{-1}G_{FB}M^{-1}G_{BF}G_{FF}^{-1} \end{pmatrix}$. Expanding $g_i^\top G^{-1} g_i$ with $g_i = (g_i^B, g_i^F)$ and collecting the four blocks gives $g_i^{F\top} G_{FF}^{-1} g_i^F + (g_i^B - G_{BF}G_{FF}^{-1}g_i^F)^\top M^{-1} (g_i^B - G_{BF}G_{FF}^{-1}g_i^F)$; since $A^\top = -G_{BF}G_{FF}^{-1}$, the second term is $\tilde{g}_i^\top M^{-1} \tilde{g}_i$. $\square$

It is a purely algebraic (Schur) identity, requiring only that G and G_{FF} be invertible (equivalently M invertible); block-diagonality of G_{FF} is not needed here, only later for the per-group decomposition. The reproduction script confirms it to 4×10^{-16} . Geometrically it is the Pythagorean split of the score into a vertical (fiber) and a horizontal (base) component under the metric-orthogonal connection A ; we use that reading only as interpretation. The vertical term summed over a group equals $1 - \pi_j$ (T3, an exact consequence of the Gelman–Pardoe definition); it is the

capacity for a group's coordinate to move under deletion. The horizontal term is the influence propagated into the hyperparameter posterior.

The next proposition justifies the name Rao–Blackwellised. It is a variance-reduction statement, stated against the correct baseline: the full-marginal importance-sampling estimator of the same marginal LOO predictive, not the conditional estimand.

Proposition 2 (Rao–Blackwell variance reduction). *Fix observation i . The PSIS-LOO importance weight is $W_i(\theta, \alpha) = 1/p(y_i \mid \theta, \alpha_{j(i)})$, and under the full-data posterior $\mathbb{E}[W_i] = 1/p(y_i \mid y_{-i})$, the reciprocal of the marginal LOO predictive (the standard case-deletion identity). Let $\bar{W}_i(\theta) = \mathbb{E}[W_i \mid \theta] = 1/p(y_i \mid \theta, y_{-i})$ be the base-conditional expectation, which is the integrand RB-LOO forms by marginalising α . Both W_i and $\bar{W}_i$ are unbiased for $\mathbb{E}[W_i]$, and, with all expectations and variances taken under the full-data posterior,*

$$\text{Var}(\bar{W}_i) \leq \text{Var}(W_i),$$

by the Rao–Blackwell (conditional-Jensen) inequality, with equality iff W_i is θ -measurable.

Proof. $\mathbb{E}[\bar{W}_i] = \mathbb{E}[\mathbb{E}[W_i \mid \theta]] = \mathbb{E}[W_i]$ by the tower rule, and $\text{Var}(W_i) = \text{Var}(\mathbb{E}[W_i \mid \theta]) + \mathbb{E}[\text{Var}(W_i \mid \theta)] \geq \text{Var}(\bar{W}_i)$, with equality iff $\text{Var}(W_i \mid \theta) = 0$ almost surely. $\square$

The inequality is useful when the excess variance lives in the fiber. If the small-group random effect makes $\text{Var}(W_i) = \infty$ while the base is well identified, so that $\text{Var}(\bar{W}_i) = \text{Var}(\mathbb{E}[W_i \mid \theta]) < \infty$, marginalising replaces an infinite-variance importance-sampling problem with a finite-variance one. That finiteness is a condition on base identifiability, not a corollary of the inequality, which otherwise reads $\infty \leq \infty$. The reciprocal-and-log step to elpd inherits the usual self-normalised consistency, not unbiasedness, and PSIS smoothing sits on top of either estimator. Empirically (Section 5) the marginalised weights have lighter tails and low base- $\hat{k}$ where the conditional weights fail.

Proposition 3 (The residual-strain predictor, T2). *To first order (one-step case deletion), the variance of the log RB-LOO base importance weight for deleting observation i , under the base posterior with covariance M^{-1} , is $\tilde{g}_i^\top M^{-1} \tilde{g}_i$, the horizontal (base) leverage. Hence the base leverage is an approximate, closed-form predictor of the residual base- $\hat{k}$ after Rao–Blackwellisation.*

Empirically the one-step base influence predicts the actual σ_u leave-one-out shift at correlation 0.987 and the residual base- $\hat{k}$ at Spearman 0.62–0.71 in the strained regime; being first-order, it under-shoots the two largest shifts, so we use it as a predictor and diagnostic, not an exact identity. As a practical flag the residual base- $\hat{k}$ itself is used directly, and Section 5 shows either quantity catches the folds where RB-LOO diverges from an exact refit (AUC 0.90–0.93).

Worked example (Gaussian random intercept). With base $(\mu, \tau = \log \sigma_u)$, fiber α, σ known, and residual $r_i = y_i - \mu - \alpha_j$, the direct base score has zero τ -component (an observation carries no first-order information about σ_u on its own), yet its leave-one-out influence on σ_u is real and flows through the connection:

$$\tilde{g}_{i,\tau} \propto r_i \alpha_j / (\sigma_u^2 G_{FF,j}),$$

proportional to the residual times the group effect, amplified by $M_{\tau\tau}^{-1} \sim 1/J$ for few groups. This is the singleton/outlier-group pathology, derived rather than assumed.

Proposition 4 (RB-LOO regimes, T4). *Conditional on the base draw, the RB-LOO integrand is: analytically exact for a Gaussian random-effect block (the rank-1 downdate of Section 4); numerically exact for a scalar Bernoulli, binomial or Poisson random-intercept block, where the*

one-dimensional integral over α_j is evaluated by deterministic quadrature to grid resolution (independent of any auxiliary-variable representation); and approximate for a non-linear block (a smooth change-point, a GP length-scale), with error controlled by the non-Gaussianity of the conditional and degrading predictably (Section 5.6). The remaining Monte Carlo error, in every case, is the base importance sampling, diagnosed by the base- $\hat{k}$ and bounded by Proposition 3.

Demarcation. The structural leverage of Theorem 1’s vertical term is Lovison et al. [2026]’s subject hat value; we do not claim it. What is not in that literature is (i) the case-deletion influence on the variance components (the horizontal term, which lies outside the variance-components-fixed hat matrix) and (ii) the cross-validation application: the pooling factor as a PSIS-reliability predictor and the base leverage as an RB-residual predictor. We claim the CV application and the composition, and cite Lovison et al. [2026], Cui et al. [2010] and Wei et al. [1998] to bound it.

4 Method: pooling-factor triage and RB-LOO

Triage (C1). Compute the pooling factor π_j and the per-observation structural leverage h_i from the Fisher blocks. Two regimes differ. In a Gaussian LMM the per-observation Fisher information is the constant $1/\sigma^2$, so h_i depends only on group size (and, through σ_u , a single fitted scalar): the ranking of folds is a design quantity, available before the outcomes are seen. In a GLMM the Fisher weight I_i depends on the fitted mean, so h_i is evaluated at the posterior mean of one model fit; it is still formed without importance weights and without case deletion, but it is post-fit, not pre-data. The triage is thus design-time relative to the cross-validation, not before fitting. Folds with high h_i are flagged as likely PSIS-LOO failures. Where group sizes are constant (as in both real-data models of Section 5.3) the group-size content of h_i is degenerate and the flag’s discrimination comes from the fitted-mean content, a point we return to in Section 5.3.

RB-LOO (C2). For each observation i in group j and each base draw $\phi^{(s)} = (\beta^{(s)}, \sigma_u^{(s)}, \dots)$, form the RE-marginalised leave-one-out predictive

$$L_{\text{RB},i}^{(s)} = \log \int p(y_i \mid \alpha, \phi^{(s)}) p(\alpha \mid y_{j \setminus i}, \phi^{(s)}) d\alpha,$$

and feed the matrix L_{RB} to `loo` for the base-only PSIS estimate. For a **Gaussian** block the integral is an analytic rank-1 downdate: with $s^2 = \sigma^2$, downdated group precision $P_{-i} = 1/\sigma_u^2 + (n_j - 1)/s^2$ and downdated mean $\hat{\alpha}_{-i} = P_{-i}^{-1} \sum_{o \in j \setminus i} (y_o - \eta_o)/s^2$,

$$L_{\text{RB},i}^{(s)} = \log \mathcal{N}(y_i \mid \eta_i + \hat{\alpha}_{-i}, s^2 + P_{-i}^{-1}).$$

For **Bernoulli, binomial and Poisson** blocks the 1-D integral over $\alpha_j \sim \mathcal{N}(0, \sigma_u^2)$ is evaluated by a self-normalised quadrature on a grid rescaled per draw by $\sigma_u^{(s)}$ (the shipped package uses 64 nodes over $\pm 6\sigma_u$ by default, widened for singleton observation-level effects with extreme responses), at negligible cost and exact to grid resolution. Because α_j is integrated out rather than importance-reweighted, the fiber-driven heavy tail is gone and the residual importance sampling is over the well-behaved base. The real-data results (Section 5.3) call the shipped `rb_loo(fit)`; the simulation studies use a reference implementation of the same estimator, which agrees with it up to the quadrature grid.

Refit flag (C3). Folds whose residual base- $\hat{k}$ still exceeds 0.7 after Rao–Blackwellisation are sent to an exact refit. For the single random-intercept GLMMs of Sections 5.2–5.3 this set is empty (one deletion barely moves a well-identified base), so RB-LOO alone suffices. In the few-groups, poorly-identified-base regime the set is non-empty and the flag earns its place: Section 5.5 shows the residual base- $\hat{k}$ catches, with AUC 0.93, exactly the folds where RB-LOO diverges from an exact refit, so the two-level triage recovers the exact answer while refitting only a few percent of folds.

Software. The `rbloo` package exposes this as a single call `rbloo(fit)`, an S3 method over `brmsfit` / `stanreg` returning, per observation, the pooling factor and structural leverage (a-priori triage), the RB-LOO pointwise elpd and base $\hat{k}$, and the refit flag. It targets the `loo` ecosystem as a drop-in.

5 Experiments

All simulations are replicated and reported with Monte Carlo standard error (MCSE); point estimates from single datasets are noisy at the group sizes that produce PSIS failures. Software versions: R 4.6.0, `loo` 2.9.0, `brms` 2.23.0 with `cmdstan/rstan`; the Gaussian case uses a hand-written conjugate Gibbs sampler.

What “matches exact refit” does and does not show. On a singleton fold RB-LOO and `reloo` compute the same marginal integral (RB-LOO by quadrature over the random-effect prior, `reloo` by refitting and resampling the dropped level from that same prior), so agreement there confirms that RB-LOO computes the intended quantity accurately but cannot by itself falsify it. The informative comparison on those folds is against PSIS-LOO and moment matching, which target the same quantity and get it wrong. The independent test, where `reloo` can falsify RB-LOO because RB-LOO’s own base importance sampling is an approximation, is the few-groups experiment of Section 5.5, where RB-LOO diverges from the exact refit on the folds its diagnostic flags. The singleton experiments establish that RB-LOO beats the importance-sampling alternatives; the few-groups experiment establishes when RB-LOO itself must defer to a refit.

5.1 Simulation studies

Two replicated studies establish C1 and C2, one conjugate and one not. The Gaussian random-intercept LMM (60 replicates of singleton-heavy designs, $\sigma_u = 1.3$; 2880 folds, 128 PSIS failures; conjugate Gibbs, with a brute-force refit gold standard on a leverage-stratified subset) is the design-time case, where the per-observation Fisher information is constant and structural leverage reduces to group size. The logistic GLMM (25 replicates, 50 groups of heterogeneous size; 2036 folds, 191 failures; MCMC fits, `reloo` gold) is the post-fit case, where the leverage’s group-size content still contributes because group sizes vary. Structural leverage separates the $k_i > 0.7$ folds a priori (AUC 0.96 and 0.81), and RB-LOO cures every failure at zero refit cost, with elpd RMSE against exact refit well below PSIS-LOO on the high- $\hat{k}$ folds (Table 1; Figure 1 shows the Gaussian case). We report per-replicate Spearman with MCSE as primary, since folds within a fit are not independent and the pooled AUC is optimistic about effective sample size.

Table 1: Simulation studies. AUC is pooled over folds; Spearman is per-replicate (mean $\pm$ MCSE). RMSE is elpd against exact refit on the high- $\hat{k}$ folds.

study	failures	C1 Spearman	C1 AUC	cured	RMSE (RB / PSIS)
Gaussian LMM (60 reps)	128	0.75 ± 0.01	0.96	100%	0.096/0.232
Logistic GLMM (25 reps)	191	0.61 ± 0.03	0.81	100%	0.015/0.060

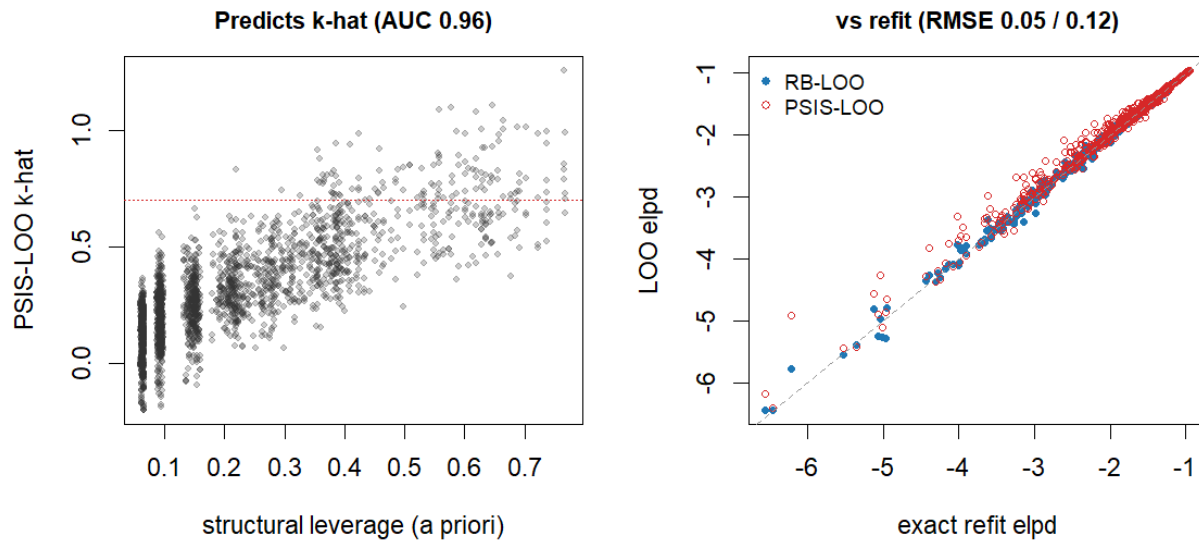


Figure 1: Gaussian LMM. Left: a-priori structural leverage predicts PSIS-LOO $\hat{k}$ (AUC 0.96). Right: RB-LOO and PSIS-LOO elpd against the brute-force refit, RMSE 0.050 and 0.123.

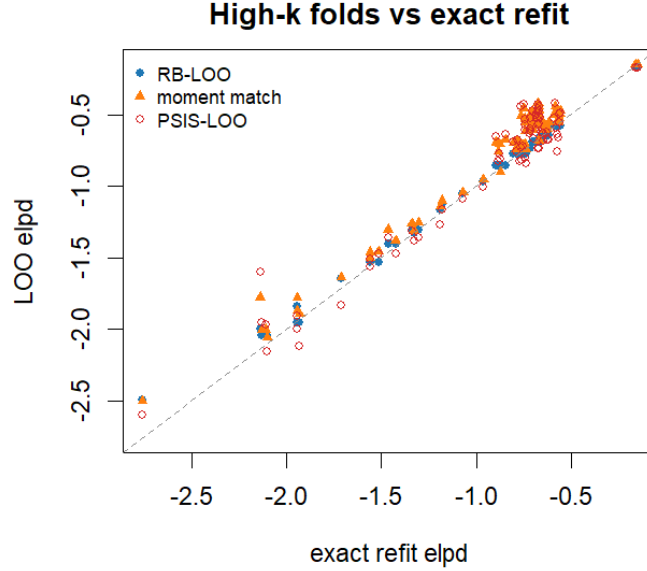


Figure 2: Moment-matching head-to-head (E-MM), logistic GLMM. On the high- $\hat{k}$ folds, RB-LOO (blue) sits on the exact-refit diagonal while moment matching (orange) and PSIS-LOO (red) scatter above it.

5.2 Moment-matching head-to-head (E-MM)

We refit the logistic GLMM design in `brms` (six replicate datasets, 77 high- $\hat{k}$ folds) and compared PSIS-LOO, moment matching [Paananen et al., 2021], and RB-LOO on the high- $\hat{k}$ folds against `reloo` gold. Moment matching performs one optimisation per high- $\hat{k}$ fold; RB-LOO performs none. Per-replicate elpd RMSE against the exact refit (mean $\pm$ MCSE) is 0.094 ± 0.025 for PSIS-LOO, 0.081 ± 0.026 for moment matching, and 0.032 ± 0.008 for RB-LOO, so RB-LOO is about $2.5\times$ more accurate than moment matching (pooled over folds, $3\times$: 0.140, 0.131, 0.044). Both RB-LOO and moment matching drive every $\hat{k}_i$ below 0.7; the difference is accuracy and cost. Moment matching barely improves on PSIS-LOO (0.131 vs 0.140) because an affine transformation of the conditional draws cannot represent a data-driven singleton random effect reverting to its prior under deletion, which is what RB-LOO marginalises in closed form. Figure 2 shows the high- $\hat{k}$ folds against the exact refit.

5.3 Real data: epilepsy (E-real)

The `brms::epilepsy` data holds 236 seizure counts on 59 patients over four visits, overdispersed and a known PSIS-LOO stressor. A patient-effect Poisson model, `count ~ zBase * Trt + zAge + (1 | patient)` evaluated with one `rb_loo(fit)` call, produces only five PSIS failures (four visits per patient). RB-LOO cures them and is the most accurate estimator against `reloo` (elpd RMSE 0.72, versus PSIS-LOO 1.76 and moment matching 0.77), but with constant group size the leverage’s discrimination is fitted-mean driven rather than group-size driven (AUC 0.84); the design-time claim is carried by the simulations of Section 5.1. The observation-level model below is the real stressor.

The overdispersion stressor. Overdispersion in count data is standardly modelled by an observation-level random effect (a Poisson-lognormal), `count ~ zBase * Trt + zAge + (1 | obs)`,

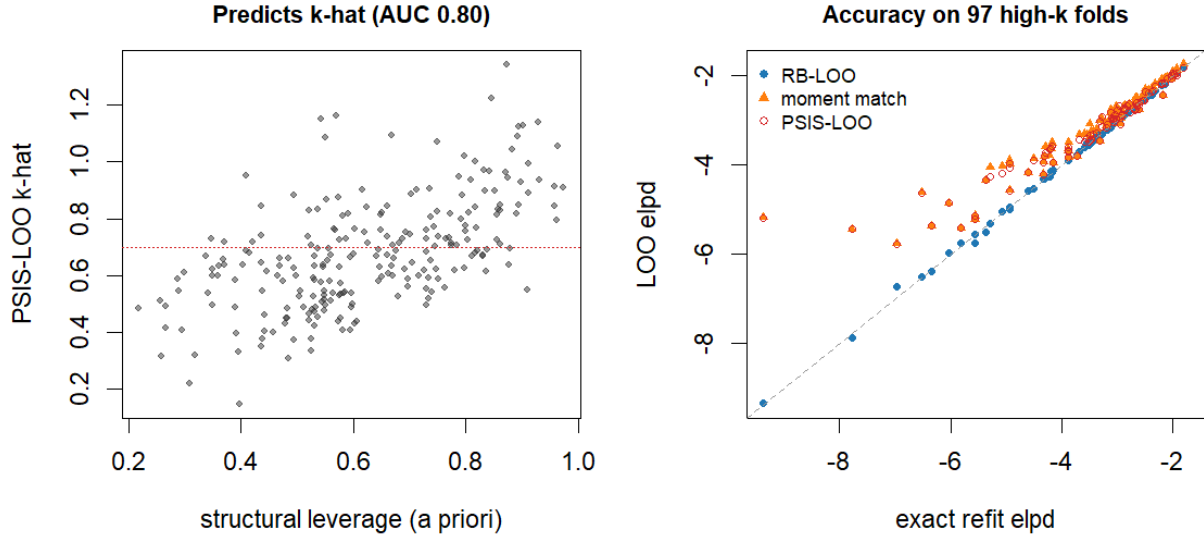


Figure 3: Epilepsy, observation-level overdispersion model. Left: structural leverage predicts PSIS-LOO $\hat{k}$ (AUC 0.80) across 97 failures. Right: against the exact refit, RB-LOO (blue) matches while PSIS-LOO (red) and moment matching (orange) are over-optimistic, most on the extreme-count folds (RMSE 0.041 against 0.638 and 0.658).

in which every observation is its own singleton group: the maximally data-driven regime. Here PSIS-LOO fails on **97 of 236** folds (max $\hat{k} = 1.34$).

- **C1.** Structural leverage predicts the failures with Spearman 0.60 and AUC 0.80 (with this many failures $\hat{k}$ is no longer floored, so the rank correlation is strong too). The groups are again all singletons, so the predictor is monotone in the fitted count (large counts are influential) rather than group-size driven.
- **C2.** RB-LOO cures all 97 (mean $\hat{k}$ 0.68 $\rightarrow$ -0.01; max 1.34 $\rightarrow$ 0.35). Moment matching leaves 37 of the 97 folds with $\hat{k} > 0.7$; it does not fix the singleton-driven failures.
- **Accuracy** against `reloo` on all 97 high- $\hat{k}$ folds (elpd RMSE): PSIS-LOO 0.638, moment matching 0.658, RB-LOO **0.041**, so RB-LOO is about 15 $\times$ more accurate than either and moment matching is here no better than raw PSIS-LOO. RB-LOO matches the exact refit to within 0.04 nats; PSIS-LOO and moment matching are systematically over-optimistic, most on the extreme-count folds (Figure 3, right).
- **Cost.** RB-LOO performs 0 optimisations; moment matching performs 97 (8 min); `reloo` performs 97 exact refits (82 minutes) for the answer RB-LOO returns at once.

On a recognised PSIS-LOO stressor, then, the state-of-the-art importance-sampling cure misses a third of the failures and is no more accurate than the incumbent, while RB-LOO reproduces the exact refit at no cost (Figure 3).

5.4 The over-optimism changes the model-selection verdict (E-decision)

Does the elpd error matter for a decision? We compare two standard models for the same overdispersed counts: M_1 , the observation-level-effect Poisson-lognormal above, and M_2 , a negative-binomial regression (`count ~ zBase * Trt + zAge, negbinomial`), which captures overdispersion

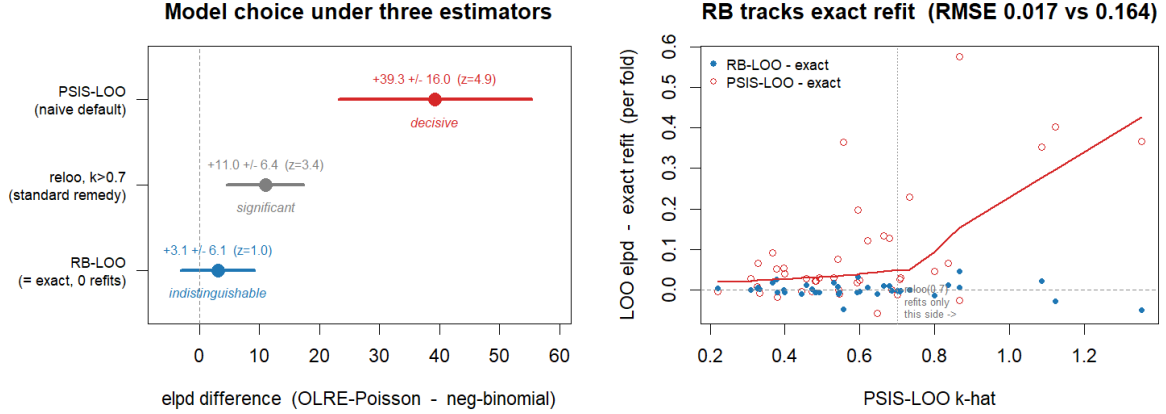


Figure 4: Model selection under three estimators (E-decision), observation-level Poisson against negative-binomial on epilepsy. Left: PSIS-LOO calls the difference decisive ($z = 4.9$), `reloo` at threshold 0.7 significant ($z = 3.4$), RB-LOO indistinguishable ($z = 1.0$). Right: against 40 exact refits, RB-LOO (blue) is unbiased across the $\hat{k}$ range while PSIS-LOO (red) is over-optimistic, biased even below 0.7 where `reloo` does not refit.

without a per-observation latent. These are near-equivalent predictive models, but M_1 's per-observation random effect makes PSIS-LOO fail on 97 of its folds while M_2 's PSIS-LOO is clean (no $\hat{k} > 0.7$). Their elpd difference is

- PSIS-LOO (naive default): $\Delta\text{elpd} = +39.3 \pm 8.0$ ($z = 4.9$), M_1 decisively preferred;
- `reloo` at the recommended $\hat{k} > 0.7$ threshold: $\Delta\text{elpd} = +11.0 \pm 3.2$ ($z = 3.4$), M_1 still significantly preferred;
- RB-LOO (zero refits): $\Delta\text{elpd} = +3.1 \pm 3.1$ ($z = 1.0$), the two models indistinguishable.

The `loo_compare` standard error $\sqrt{N} \text{sd}(d_i)$ is itself uncertain and mildly anti-conservative [Sivula et al., 2020], so we read the z -values as descriptive; the robust finding is the shift $39.3 \rightarrow 11.0 \rightarrow 3.1$, which does not depend on the standard error. To adjudicate, we exact-refit 40 folds stratified across the $\hat{k}$ range. RB-LOO matches the exact refit fold-by-fold (RMSE 0.017, mean bias +0.001), whereas PSIS-LOO is over-optimistic with a bias that grows in $\hat{k}$ (mean +0.088; +0.207 on the $\hat{k} > 0.7$ folds), and is already biased on the 0.55–0.7 folds (+0.092) that `reloo` at threshold 0.7 does not refit. The standard `reloo` remedy therefore inherits residual PSIS over-optimism on its sub-threshold folds; its +11.0 is itself optimistic, and RB-LOO's +3.1, quadrature-converged (stable to 0.1 nat across grids from $\pm 6\sigma$ to $\pm 16\sigma$) and validated fold-wise against exact refits, is the correct verdict. The comparison is consistent: each model's elpd is computed by a method accurate for that model. M_1 's failing per-observation effect is what RB-LOO corrects (validated to the exact refit); M_2 has no random effect, its PSIS-LOO is reliable (max $\hat{k} = 0.51$), and exact-refitting all five of its $\hat{k} > 0.2$ folds moves its total by 0.0 nats, so M_2 's elpd is itself exact. RB-LOO is applied only where the error is. The asymmetry is intrinsic: PSIS-LOO's over-optimism biases model comparison when the candidates differ in latent structure (here, a latent-variable model against a marginal one), where the per-model bias does not cancel in the difference. A practitioner using default PSIS-LOO, or the recommended `reloo` threshold, would report decisive or significant evidence for the more complex model where an exact analysis finds none (Figure 4).

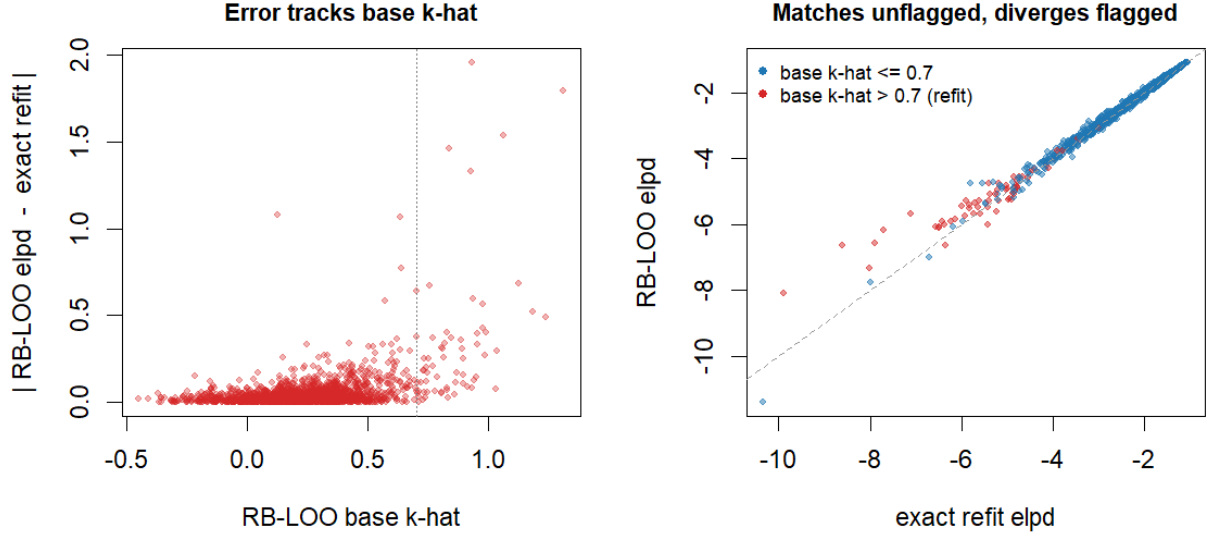


Figure 5: RB-LOO’s own limit and the refit flag (E-basestress), few-groups Gaussian LMMs. Left: RB-LOO’s elpd error against the exact refit grows with its own residual base- $\hat{k}$. Right: RB-LOO matches the exact refit on the unflagged folds (blue) and diverges on the flagged folds (red), which the flag catches.

5.5 When RB-LOO itself fails, and the refit flag (E-basestress)

The experiments above stress PSIS-LOO; this one stresses RB-LOO. We use few-groups Gaussian LMMs ($J = 4$ to 6 groups with singletons, weakly-identified σ_u ; 2340 folds over replicated designs), where deleting one observation moves the base posterior appreciably, so RB-LOO’s residual base importance sampling is a real approximation and a brute-force refit of every fold is an independent oracle that can falsify it.

- RB-LOO’s residual base- $\hat{k}$ exceeds 0.7 on 65 folds, so RB-LOO is strained here, unlike the singleton experiments.
- RB-LOO is not exact on the flagged folds: elpd RMSE against the exact refit is 0.068 on the unflagged folds (base- $\hat{k} \leq 0.7$) but 0.532 on the flagged folds (base- $\hat{k} > 0.7$), so RB-LOO’s own diagnostic tracks its own error.
- The residual base- $\hat{k}$ separates the folds where RB-LOO diverges from the exact refit (by more than 0.25 nats) with AUC 0.93; the base leverage (T2) does so at AUC 0.90.
- Applying RB-LOO on the unflagged folds and an exact refit on the flagged folds gives elpd RMSE 0.067 against exact, refitting only 3% of folds, against 0.111 for RB-LOO alone and 0.389 for PSIS-LOO.

Here `reloo` does reveal RB-LOO error, where the flag predicts, and the two-level triage delivers refit-quality LOO at a few percent of the refit cost (Figure 5).

5.6 The linearisation boundary (E-M5)

Beyond the base importance sampling, the other approximation is the RB integrand itself, which is analytic for Gaussian blocks and numerically exact for the scalar GLMM families but only

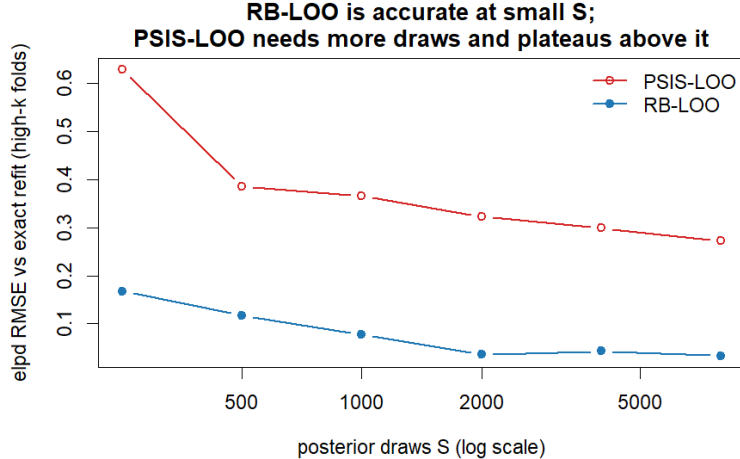


Figure 6: Efficiency in the number of draws (E-Scaling). On the high- $\hat{k}$ folds, RB-LOO reaches refit-quality accuracy with far fewer posterior draws than PSIS-LOO, which plateaus above it.

approximate for non-linear blocks. In a smooth logistic change-point model (the `bjlm` setting) we mapped where the linearised conditional degrades. A moment-matched Gaussian conditional stays calibrated across sparsity and sharpness (Kolmogorov–Smirnov statistic 0.018 to 0.043), and the raw Fisher/pooling approximation is adequate except at extreme change-point sharpness ($\rho = 20$: KS 0.060 against the 0.048 critical value), where the mode $\neq$ mean gap and the dropped curvature bite. The degradation is monotone and predictable, which is the limit of the method for non-conjugate blocks.

5.7 Efficiency in the number of draws (E-Scaling)

The “zero refit cost” claim concerns model refits; a second efficiency axis is the number of posterior draws S . On a failing fold PSIS-LOO’s importance weights have (near-)infinite variance, so its elpd estimate is noisy and biased and improves only slowly in S ; RB-LOO marginalises the offending direction, so its weights have finite variance and its estimate converges quickly. On the same Gaussian LMM design, drawing one long conjugate chain and subsampling its first S draws (so only S varies), we measure elpd RMSE against exact refit on the high- $\hat{k}$ folds over $S \in \{250, \dots, 8000\}$ (Figure 6). RB-LOO at $S = 250$ (RMSE 0.17) is already more accurate than PSIS-LOO at $S = 8000$ (RMSE 0.27), roughly an order of magnitude fewer draws, and over this range PSIS-LOO improves far more slowly and plateaus well above RB-LOO. PSIS-LOO remains consistent as $S \rightarrow \infty$, but its convergence on infinite-variance folds is slow enough that in practice it does not reach RB-LOO’s accuracy.

6 Discussion

RB-LOO applies to a single random-intercept grouping factor with a scalar random-effect block: Gaussian LMMs (analytic) and Bernoulli, binomial and Poisson GLMMs (1-D quadrature). Its estimand is the integrated (marginal-over-RE) predictive, against which we compare throughout [Merkle et al., 2019]. Random slopes and correlated within-group effects are a low-dimensional extension, where the quadrature becomes two- or three-dimensional but stays cheap. Crossed or multiple grouping factors are the hard case: a deleted observation loads on several fibers at once

and there is no single block to collapse, so the package falls back to plain PSIS-LOO, which we name as out of scope rather than degrade silently. Practically, the method is a two-line change to a hierarchical LOO workflow: triage on the pooling factor, integrate out the flagged folds, and refit only the residual few, which removes most of the refits PSIS-LOO would otherwise demand. The geometry is interpretation: the pooling factor is the vertical half of a case-deletion influence identity whose horizontal half is the variance-component funnel, and cross-validation reliability reads off the two halves.

7 Honesty and limitations

- **The marginalisation cure is not novel.** It is integrated importance-sampling LOO [Merkle et al., 2019, Vehtari et al., 2016, Bürkner et al., 2021], the standing `loo` recommendation for hierarchical PSIS failures, and the same estimand as INLA’s CPO [Rue et al., 2009, Held et al., 2010]. We claim the a-priori triage, the packaged closed forms, the head-to-head, and the geometry, not the marginalisation.
- **The “a-priori / design-time” claim is exact only for the Gaussian case** (constant Fisher information $\Rightarrow$ leverage = group size). For GLMMs the predictor uses one model fit (no importance weights, no case-deletion); on constant-group-size real data (both epilepsy models) its discrimination is fitted-mean-driven, closer to “influential points have high $\hat{k}$ ” folklore than to a pure group-size claim. The group-size content is demonstrated on the heterogeneous-group simulations.
- **“Matches exact refit” is estimand-shared on singleton folds**, where `reloo` and RB-LOO compute the same integral, so it cannot falsify RB-LOO there; the meaningful comparison on those folds is against PSIS-LOO and moment matching. The independent falsification is E-basestress (Section 5.5), where RB-LOO does err and the flag catches it.
- **Pooled figures over-count.** Folds within a fit are not independent; we report per-replicate statistics with MCSE as primary (e.g. E-MM per-replicate $2.5\times$ vs the pooled $3.0\times$) and flag pooled AUCs as optimistic about effective sample size.
- RB-LOO targets the integrated predictive; every accuracy comparison uses a refit / `reloo` with the same target. The integrand is analytic (Gaussian), numerically exact (scalar GLMM quadrature), or approximate (non-linear blocks), mapped in Section 5.6.
- The moment-matching comparator used the recommended robust split transformation at default settings; we did not sweep its tuning, and we did not run mixture importance sampling [Silva and Zanella, 2024], an alternative cure we discuss but do not benchmark.
- The base leverage is the theoretically-correct base-IS-failure predictor but is empirically close to the pooling factor / residual base- $\hat{k}$ for flagging RB residual failures (AUC 0.90 vs 0.93); we present it as explanation, with the residual base- $\hat{k}$ as the operational flag.

Reproducibility

All results are produced by the scripts listed below (in `analysis/` in the repository; R 4.6.0, `loo` 2.9.0, `brms` 2.23.0 with `rstan` 2.32.7 / `cmdstan` 2.38.0; GLMM fits use 4 chains of 1000–2000 iterations; seeds are fixed in each script). The RB-LOO estimator is implemented once in the

`rbloo` package (`rb_loo()`); the real-data results call it directly, and the simulation studies use a self-contained reference implementation of the identical estimator (agreeing up to the quadrature grid).

Result	Float	Script
Gaussian LMM	Fig. 1, Tab. 1	<code>rb_loo.R</code>
Logistic GLMM, replicated	Tab. 1	<code>replicate_glmm.R</code>
Moment-matching head-to-head (E-MM)	Fig. 2	<code>emm_moment_match.R</code>
Real data, patient effect	§5.3	<code>ereal_epilepsy.R</code>
Real data, overdispersion stressor	Fig. 3	<code>ereal_epilepsy_olre.R</code>
Model selection (E-decision)	Fig. 4	<code>e_decision.R</code> , <code>verify_m2_full.R</code> , <code>adjudicate.R</code>
RB-LOO’s own limit (E-basestress)	Fig. 5	<code>e_basestress.R</code>
Linearisation boundary (E-M5)	§5.6	<code>bjlm_changepoint_rb.R</code>
Draw-count efficiency (E-Sscaling)	Fig. 6	<code>s_scaling.R</code>

Code and data availability. The `rbloo` package and all experiment scripts are available at <https://github.com/ABindoff/rbloo>. The real datasets are public: `epilepsy` ships with `brms`. No new data were collected.

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